Valecium OS

Figure 1: ValeciumOS Icon

Valecium OS is a Unix-like operating system project focused on low-level systems design, modular kernel architecture, and educational clarity.

This repository contains:

- A kernel with architecture-specific and architecture-agnostic layers
- Filesystem, memory, process, and syscall subsystems
- Small userspace components (`libmath`, `sh`)
- Build tooling around SCons, QEMU, and a managed cross-toolchain

Documentation

Primary source docs live in `Documents/`.

- Project index: `Documents/index.ad`
- Build guide: `Documents/building.ad`
- Development guide and coding conventions: `Documents/devel.ad`
- Implementation roadmap: `Documents/roadmap.ad`

Published docs are available at docs.vyang.org/valecium.

Quick Start

1. Install host prerequisites

You need Python 3 and SCons.

Ubuntu example:

```
sudo apt install python3 scons
```

2. Install dependencies

```
scons deps
```

3. Build

```
scons
```

4. Run or debug

```
scons run
# or
scons debug
```

Notes:

- Build settings are persisted in `.config` at repository root (auto-created on first run).
- Toolchain setup is handled by the build flow (`--ensure`) and is also available explicitly via `scons toolchain`.
- On macOS, image-building support is limited (notably due to `parted` availability). See `Documents/building.ad`.

Repository Guide

Top-level directories and what they contain:

- `kernel/` - kernel source: CPU, memory, drivers, filesystems, syscalls, HAL, init
- `include/` - public and internal headers used by kernel/userspace
- `usr/` - userspace components and libraries
- `image/` - disk image assembly logic and root image content

- `scripts/` - helper scripts for dependencies, toolchain, QEMU, GDB, formatting
- `Documents/` - AsciiDoc source for project documentation

Important build files:

- `SConstruct` - root build entrypoint and global configuration
- `kernel/SConscript` - kernel build rules
- `usr/SConscript` - userspace build rules
- `image/SConscript` - image build rules

Build Configuration Reference

Common SCons variables:

- `config`: `debug` (default) or `release`
- `arch`: `i686` (default), `x64`, `aarch64`
- `buildType`: `full` (default), `kernel`, `usr`, `image`
- `imageFS`: `fat12`, `fat16`, `fat32` (default), `ext2`
- `imageSize`: image size (for example `250m`)
- `outputFormat`: `img` (default) or `iso`
- `outputFile`: base name of output image

Examples:

```
scons config=release
scons arch=x64 config=release
scons buildType=kernel
scons outputFormat=iso
```

For full build and platform notes, read `Documents/building.ad`.

Development and Contributing

Start with `Documents/devel.ad`, which covers:

- Build/development workflow
- Git and patch submission expectations
- Coding style, naming, and commenting conventions
- Architecture separation and HAL strategy (no platform-specific assembly in common code)

If you plan a larger feature, check `Documents/roadmap.ad` to align with current priorities.

Project Status

Valecium OS is actively developed. Roadmap highlights include:

- Core kernel infrastructure improvements

- Expanded syscall coverage
- Filesystem and device capability growth
- Longer-term advanced features (networking, extended process model)

Track current progress in `Documents/roadmap.ad`.

Licensing

Project licensing summary is in `COPYING`. Individual license texts are in `LICENCES/`.

Support

If you find a bug or want to request a feature, open an issue on the GitHub repository and include:

- Target architecture/config
- Build command used
- Relevant logs or reproduction steps